

Chapter 130

Camera Raw Palette

The Camera Raw palette lets you make clip-specific adjustments to the parameters that are used to debayer a raw clip into an image that can be graded in DaVinci Resolve.

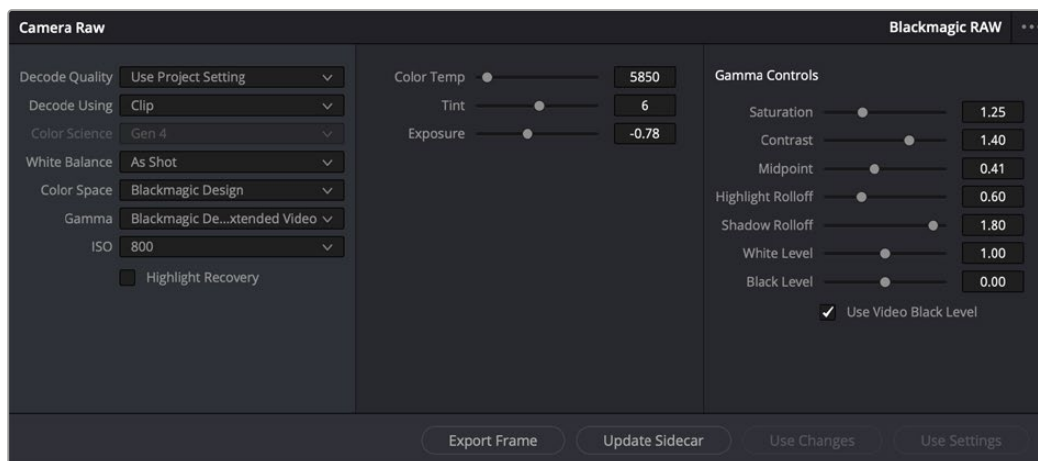
Contents

Introduction to the Camera Raw Palette.....	3053
Copying, Versioning, and Protecting Camera Raw Settings.....	3053
Making Changes to Clip Camera Raw Settings.....	3054
Clip Decoder Settings.....	3055
Resetting Camera Raw Settings.....	3056
Updating Sidecar Settings for Blackmagic RAW (BRAW) Clips.....	3056

Introduction to the Camera Raw Palette

When a timeline uses clips that are linked to camera raw source media recorded from cameras from Blackmagic Design, RED, ARRI, Sony, and Vision Research, all clips in raw media formats are initially debayered using the settings found in the Camera Raw panel of the Project Settings.

However, if there are individual clips that you want to apply different raw settings to, for example altering the ISO to pull more detail out of the highlights or shadows, then you can use the controls found in the Camera Raw palette to individually alter the parameters found within.



The Camera Raw palette showing the available parameters for Blackmagic RAW media.

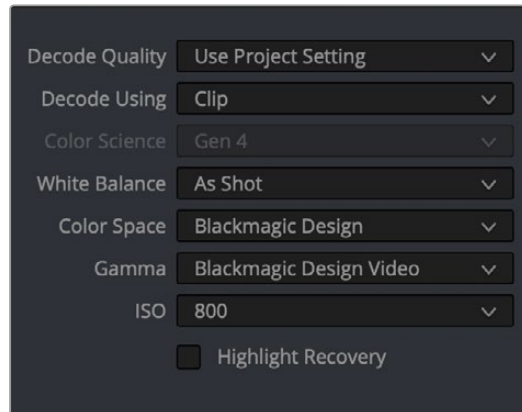
The Camera Raw palette is automatically set to the mode (seen within the Mode drop-down menu) that is appropriate to the clip that's currently selected. If the current clip is not in a raw format, then the parameters within the Camera Raw palette are disabled.

This section covers general use of the Camera Raw palette. For in-depth documentation about specific camera raw parameters, see *Chapter 7, "Camera Raw Settings."*

Copying, Versioning, and Protecting Camera Raw Settings

Ordinarily, a clip's camera raw settings are copied along with its grade, or saved inside stills grabbed from that clip, when you use the various grade management techniques covered in later Chapter, see *Chapter 142, "Grade Management."*

When you create new versions, you copy the current Camera Raw settings to the new version, but any changes you make are specific to that version, so each version can have individual Camera Raw adjustments. For example, you could compare the results of two different camera raw adjustments on the same clip.



Camera Raw master settings for BRAW media

If you're copying and rippling grades among multiple clips, you can also protect each clip's camera raw settings from being overwritten using the "Copy Grade: Preserve Camera Raw Settings" option found in the contextual menu of the Gallery.

For more information on the Copy Grade settings, see *Chapter 142, "Grade Management."*

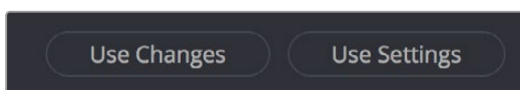
Making Changes to Clip Camera Raw Settings

If you want to make individual adjustments to a particular clip's camera raw settings, choose "Clip" from the Decode Using drop-down menu in the Camera Raw palette. This makes all the parameters in the Camera Raw palette editable, and changes you make override the project-wide camera raw settings.

Changes to the parameters in the Camera Raw palette can also be rippled across multiple clips at once.

To ripple camera raw adjustments across multiple clips:

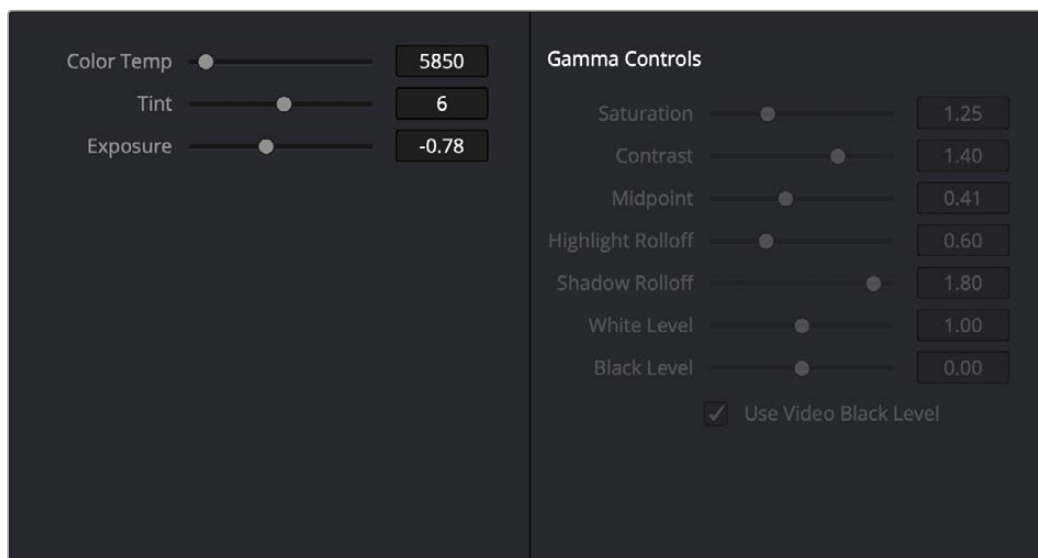
- 1 First, you must select a range of clips in the Color page timeline.
- 2 Open the Camera Raw palette, and make whatever adjustments are necessary to the current clip. The name of each parameter you adjust changes to amber, showing you which parameters have been modified, and which have not.
- 3 To ripple your changes, do one of the following:
 - Click the Use Changes button to ripple only the altered parameters (in amber) to the other clips you've selected in the Timeline. This preserves differences between clips in the parameters you haven't adjusted (in gray).
 - Click the Use Settings button to ripple every parameter of the current clip to the other clips you've selected, overwriting all the camera raw settings at once.



The Use Changes and Use Settings buttons in the Camera Raw palette

Clip Decoder Settings

There's much more information on the various format-specific Master settings, as well as the occasionally format-specific Clip Decoder settings, in Chapter 7, "Camera Raw Settings." However, with the exception of the RED Clip Decoder settings that appear for R3D clips, most other formats share a set of DaVinci Resolve-specific controls that provide wide-latitude access to the raw image data for purposes of making different kinds of adjustments.



Camera Raw Clip Decoder settings for BRAV media

While specific raw formats have individual controls, the standard controls include:

- **Color Temp:** Designed to alter the “warmth” of the image. Adjustable in degrees Kelvin. Lower values correct for “warmer” lighting, while higher values correct for “cool” lighting. +6500 is unity. The range is +2000 to +50,000.
- **Tint:** Color balance correction for images with a green or magenta color cast, such as fluorescent or sodium vapor bulbs. 0 is unity. The range is –150 to +150.
- **Exposure:** Increases or lowers image lightness in units relative to *f*-stops. If your intended exposure adjustment lifts image data above the maximum white level, don't worry; all image data is preserved and can be retrieved in subsequent adjustments. 0 is unity. The range is –5 to +5.
- **Sharpness:** A debayer-specific sharpness filter applied to provide the appearance of enhanced image detail. 20 is unity. The range is 0 to 100.
- **Highlights:** Makes it easy to selectively retrieve blown-out highlight detail in high-dynamic-range media by lowering this parameter, and achieves a smooth blend between the retrieved highlights and the unadjusted midtones for a naturalistic result. 0 is unity. The range is –100 through +100.
- **Shadows:** Lets you selectively lighten or darken shadow detail. Raising this value retrieves shadow detail recorded below 0 percent, while leaving the midtones alone. 0 is unity. The range is –100 through +100.
- **Color Boost:** A non-uniform saturation operation that affects regions of low saturation more than regions of high saturation. This is sometimes referred to as a vibrance operation. 0 is unity, showing the original color values. Raising color boost from 0-100 increases color intensity, but low-saturation parts of the image are raised more aggressively. Lowering Color Boost from 0 to -100 decreases color intensity, but low-saturation parts of the image are lowered more aggressively. 0 is unity, showing unaltered saturation. The range is –100 through +100.

- **Saturation:** A uniform saturation operation that raises (above 50) or lowers (below 50) the color intensity of every color value within the image. 50 is unity, showing unaltered saturation. The range is -100 (completely desaturated) through +100 (saturation is doubled).
- **Midtone Detail:** When this parameter is raised, the contrast of regions of the image with high edge detail is raised to increase the perception of image sharpness, sometimes referred to as definition. When this parameter is lowered to a negative value, regions of the image with low amounts of detail are softened while areas of high-detail are left alone. 0 is unity. The range is -100 through +100.
- **Lift:** Adjusts the black point of the media, raising it or lowering it while scaling all midtone values between it and the white point. Regardless of how you adjust this control, all image data is preserved and can be retrieved in subsequent adjustments. The range is -100 to +100.
- **Gain:** Adjusts the white point of the media, raising or lowering it while scaling all midtone values between it and the black point. Regardless of how you adjust this control, all image data is preserved and can be retrieved in subsequent adjustments. 0 is unity. The range is -100 to +100.
- **Contrast:** Raising contrast reduces shadows and raises highlights, while leaving midtones at 50 percent unaffected. Regardless of how you adjust this control, all image data is preserved and can be retrieved in subsequent adjustments. 0 is unity. The range is -100 to +100.

Resetting Camera Raw Settings

If you've made changes to the parameters of the Camera Raw palette and you decide you need to reset them, there are two options, found in the Options menu.

- **Reset:** Resets all parameters in the Camera Raw palette to their default settings.
- **Revert:** Similar to the "Original Memory" command, Revert changes all camera raw parameters back to the state they were at when you first selected the current clip.

Updating Sidecar Settings for Blackmagic RAW (BRAW) Clips

Blackmagic RAW clips support both embedded look metadata within the .braw media clip and external look metadata in .sidecar files. Ordinarily, these files travel in pairs whenever you manage BRAW media, and whenever a .sidecar file is present, its settings override those of the embedded metadata within the actual .braw clip. Since the .sidecar metadata is in human-readable JSON formatted text, it's easy to edit and accommodates a wide variety of onset to post-production workflows.

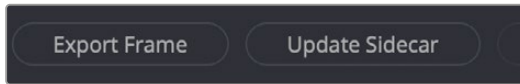


The .braw and .sidecar files as seen in the file system

If you like, you can make changes to the raw look metadata of a .braw clip and upgrade the .sidecar metadata from within the Color page of DaVinci Resolve, so the media on disk reflects these changes in any other BRAW compatible application.

To update the .sidcar raw look metadata:

- 1 Open the Color page, select a .braw clip in the Thumbnail timeline, and open the Camera Raw palette.
- 2 Set Decode Using to Clip to enable the palette controls.
- 3 Make whatever adjustments you want to make using the controls of the Camera Raw palette.
- 4 When you're done, click the Update Sidecar button.



The Update Sidecar button in the Camera Raw palette for .braw media